

ApexPlanet Data Analytics Internship — Task 4

Data Storytelling & Business Intelligence Report

Prepared for stakeholder presentation and final internship submission

Project Snapshot

Total Sales	Quantity Sold	Customers	Unique Orders
₹139.4M	5,435	947	992

1. Executive Summary

This report consolidates the key findings from the ApexPlanet Data Analytics Internship Task 4 analysis. The objective was to understand overall sales performance, identify where revenue is concentrated, evaluate customer, category, product, geographic and monthly patterns, and statistically validate an important transaction-level comparison.

The analysis covers 1,000 records across 12 columns from January 2025 to January 2026, with 0 missing values and 0 complete duplicate rows. Total sales were approximately ₹139.4 million from 5,435 units, 947 unique customers and 992 unique Order IDs. The strongest business finding is that approximately 25% of customers classified as High Value contributed approximately 55% of total sales. Electronics was the largest revenue-generating category at approximately ₹50.78 million (36.43% of total sales).

A Welch independent two-sample t-test was used to test whether Electronics transactions had a significantly different mean transaction sales value from Non-Electronics transactions. Although Electronics had a higher observed mean (₹143,442.32 versus ₹137,183.99), the difference was not statistically significant ($p = 0.413674$). Therefore, business decisions should combine descriptive business intelligence with statistical evidence rather than relying on observed differences alone.

2. Business Objective

The business objective was to answer the following question: How is the business performing, and which customers, categories, products, cities and periods drive its sales?

The analysis therefore focused on:

- Overall sales performance
- Customer contribution and value concentration
- Category and product performance
- Geographic variation
- Transaction-level behaviour
- Statistical validation
- Actionable business recommendations

3. Data Overview

The final analysis dataset contains 1,000 records and 12 columns. The analysed period runs from January 2025 to January 2026. The dataset contains 947 unique customers and 992 unique Order IDs. There are 0 missing values and 0 complete duplicate rows.

Key measures:

- Total Sales: approximately ₹139.4M
- Total Quantity Sold: 5,435 units
- Unique Customers: 947
- Unique Orders: 992
- Average Order Value: ₹140,523.63, based on unique Order IDs

January 2026 contains partial-period data and was therefore not compared directly with complete months in the monthly trend analysis.

4. Overall Sales Performance

The business generated substantial sales volume across the analysed period. Total sales reached approximately ₹139.4M from 5,435 units. The customer base was broad, with 947 distinct customers contributing to 992 distinct Order IDs across 1,000 records.

The average order value was ₹140,523.63 when calculated using unique Order IDs. This provides a useful high-level benchmark for understanding the value associated with individual orders.

5. Customer Value Concentration

Revenue is strongly concentrated among High Value customers. Approximately 25% of customers generated approximately 55% of total sales. Medium Value customers represented approximately 50% of customers and contributed approximately 41% of sales, while Low Value customers represented approximately 25% of customers and contributed approximately 4% of sales.

This concentration indicates that customer retention and relationship management are particularly important for the High Value segment. Protecting this group can have a disproportionately large effect on overall revenue stability.

6. Category and Product Performance

Electronics was the leading revenue-generating category, contributing approximately ₹50.78M or 36.43% of total sales. Education was the next-largest category.

At product level, Laptop, Mobile and Book were the leading products. Laptop generated approximately ₹25.44M, while Mobile generated approximately ₹25.34M in sales.

A key analytical distinction is important: being the largest category by total sales does not automatically mean that individual Electronics transactions have significantly higher values than transactions in other categories. This distinction is addressed through statistical validation in Section 8.

7. Geographic and Monthly Performance

Sales performance varied across cities. Patna recorded the highest total sales at approximately ₹19.29M. Bengaluru recorded the highest sales per customer at approximately ₹156,446, while Kolkata and Bengaluru were also among the strongest cities.

Monthly performance also varied. March 2025 was the highest complete month at approximately ₹13.06M, while September 2025 was the lowest complete month at approximately ₹9.18M. January 2026 was treated as a partial period and excluded from direct complete-month comparison.

These patterns can support geographic growth planning and time-based sales and inventory planning.

8. Statistical Validation

Business Question: Do Electronics transactions have a significantly different average transaction sales value compared with Non-Electronics transactions?

Null Hypothesis (H0): There is no significant difference in mean transaction sales between Electronics and Non-Electronics.

Alternative Hypothesis (H1): A significant difference exists.

Test: Welch's independent two-sample t-test

Significance level (α): 0.05

Results:

- Electronics transactions: 354
- Non-Electronics transactions: 646
- Electronics mean: ₹143,442.32
- Non-Electronics mean: ₹137,183.99
- Mean difference: ₹6,258.33
- t-statistic: 0.8179
- p-value: 0.413674
- 95% confidence interval: -₹8,764.16 to ₹21,280.82
- Cohen's d: 0.0548 (Negligible)
- Decision: Fail to Reject H0

Interpretation: Electronics transactions have a higher observed average sales value, but the evidence is not statistically strong enough to conclude that the population means are different at the 5% significance level. The confidence interval also includes zero, supporting the same conclusion.

9. Key Business Insights

Five findings connect the descriptive and statistical evidence:

1. Revenue is strongly concentrated among High Value customers: approximately 25% of customers contribute approximately 55% of total sales.
2. Electronics is the largest revenue category at approximately 36.43% of total sales.

3. Laptop and Mobile are the top products, with approximately ₹25.44M and ₹25.34M in sales respectively.
4. Sales performance varies across cities and months: Patna leads total sales, while Bengaluru leads sales per customer.
5. There is no statistically significant transaction-value difference between Electronics and Non-Electronics ($p = 0.413674$).

10. Business Recommendations

1. Prioritise High Value customer retention: Build retention and relationship programmes for the segment responsible for approximately 55% of sales.
2. Maintain Electronics strength: Keep inventory, pricing and availability aligned with demand in the largest category.
3. Monitor Laptop and Mobile: Track the two leading products closely and monitor the wider product portfolio.
4. Use geographic insights: Consider both total sales and sales per customer to identify growth opportunities across cities.
5. Monitor monthly trends: Use time-based patterns to guide sales planning and inventory decisions.
6. Pair metrics with statistics: Use statistical validation alongside descriptive metrics for important business decisions.

11. Conclusion and Call to Action

The analysis shows strong overall sales performance, with revenue concentrated among High Value customers and Electronics emerging as the leading revenue-generating category. Laptop and Mobile are the strongest products, while geographic and monthly variation provides additional opportunities for targeted planning.

However, statistical validation demonstrates that the higher observed average transaction value for Electronics is not statistically significant compared with Non-Electronics. This reinforces the importance of distinguishing descriptive performance from statistically validated differences.

Recommended next actions are to strengthen High Value customer retention, monitor Electronics and key products, identify geographic growth opportunities, track monthly sales trends, and use statistical validation when making important business decisions.

12. Analytical Note

This report is based on the final Task 4 stakeholder presentation and its underlying analytical outputs. The report preserves the reported figures and conclusions from the project. No external data has been introduced. January 2026 is explicitly treated as partial-period data for monthly comparisons.